

FEATURE SPECIFICATION

Project: Complaint Stack

1. Document Control

Attribute	Details
Document Title	Core Feature Specification & System Modules
Project Name	Complaint Stack
Document Version	1.0.0 (Baseline Requirements)
Status	Approved for Development

2. Executive Summary

Complaint Stack is a comprehensive, multi-tier platform designed to bridge the gap between citizens and municipal/departmental officers. The system facilitates secure identity management, targeted issue routing, and transparent resolution tracking. By categorizing users into Citizens (Users), Responding Officers, and System Administrators, the architecture ensures strict operational boundaries while maintaining real-time visibility into complaint lifecycles.

3. Core System Modules & Features

3.1 Module 1: User Registration & Identity Management

- **Secure Sign-Up:** Implements robust authentication protocols allowing users to register securely utilizing email and password credentials.
- **Profile Creation & Storage:** Captures and securely stores essential user demographics, including full name, verified contact information, and primary identification details. This foundational data ensures accurate historical tracking and accountability for all future complaint submissions.

3.2 Module 2: Department Routing & Officer Browsing

- **Search and Filter Capabilities:** Provides an intuitive interface for users to filter complaint categories and route issues directly to the appropriate civic departments (e.g., Police, Electricity, Municipality, Public Works).
- **Real-Time Status Tracking:** Empowers users with live visibility into officer availability and the exact deterministic state of their submitted complaints (e.g., *Pending, In Progress, Resolved*).

3.3 Module 3: Complaint Lodging & Lifecycle Management

- **Streamlined Complaint Form:** Features a frictionless submission portal where users select specific categories, provide detailed textual descriptions, and upload multimedia evidence (documents and images) to support their claims.
- **Automated Notification Engine:** Integrates an event-driven notification system that dispatches automated Email and SMS alerts at critical lifecycle milestones (submission confirmation, status transitions, and final resolution), ensuring continuous user engagement.

3.4 Module 4: Officer Resolution Dashboard

- **Assigned Queue Management:** Provides a dedicated, secure dashboard for approved officers to view, filter, and prioritize their specific workload of assigned complaints.
- **Action Logging & Evidence Uploads:** Enables officers to append official status updates, record internal action notes, and securely attach final resolution reports or photographic evidence to close out tickets.

3.5 Module 5: Administrative Control & Verification Center

- **Officer Approval System:** Acts as the primary security gateway for the responding workforce. Administrators manually verify credentials and approve officer accounts to prevent unauthorized access and maintain strict system integrity.
- **System Monitoring & Escalation:** Equips administrators with a high-level command center to monitor platform analytics, resolve user-officer disputes, handle high-priority escalations, and audit overall platform security.

4. Standard Operating Workflows

1. **User Journey:** Registration → Browse Departments → Lodge Complaint (with Attachments) → Receive SMS/Email → Track Live Status.
2. **Officer Journey:** Admin Verification → Access Dashboard → View Assigned Queue → Execute Field Work → Log Action/Report → Mark Resolved.

3. **Admin Journey:** Monitor Analytics → Verify New Officers → Audit Overdue Complaints → Enforce System Security.